

Module 3: JavaScript Fundamentals

Following the exploration of Java backend development and frontend basics, this module focuses on the engine of web interactivity: JavaScript.

1. Introduction to JavaScript

JavaScript is a high-level, interpreted programming language that enables interactive web pages. While HTML provides structure and CSS provides style, JavaScript provides behavior. It is essential for modern web applications to handle user input, communicate with servers, and update the UI dynamically.

2. Variables

Variables are containers for storing data values. In modern JavaScript, we use three main keywords to declare them:

- `let`: Declares a block-scoped variable that can be reassigned.
- `const`: Declares a block-scoped variable that cannot be reassigned (constant).
- `var`: The older way to declare variables; function-scoped and generally avoided in modern code.

Example:`let userName = "Rajesh";`

`const plantId = 101;`

`userName = "Nakkana"; // Allowed`

`// plantId = 102; // Error: Assignment to constant variable`

3. Data Types

JavaScript is dynamically typed, meaning a variable can hold different types of data at different times.

Type	Description	Example
String	Textual data wrapped in quotes	"BHEL"
Number	Integers or floating-point numbers	45, 3.14
Boolean	Logical values	true, false
Object	Complex data structures	{ id: 1, name: "Unit A" }
Undefined	A variable that has not been assigned a value	let x;
Null	Represents the intentional absence of any object value	let y = null;

4. Operators

Operators allow us to perform calculations and comparisons.

- **Arithmetic:** + (Addition), - (Subtraction), * (Multiplication), / (Division), % (Modulus).
- **Assignment:** =, +=, -=.
- **Comparison:** == (equal to), === (equal value and type), != (not equal), > (greater than).
- **Logical:** && (AND), || (OR), ! (NOT).

5. Conditional Statements

Conditionals control the flow of the program based on different conditions.

If - Else

Used to execute a block of code if a condition is true, and another block if it is false.

```
let temperature = 35;
```

```
if (temperature > 30) {
```

```
    console.log("Cooling system activated.");
```

```
} else {
```

```
    console.log("Normal operating temperature.");
```

```
}
```

Switch

Used to perform different actions based on different conditions (multiple cases).

```
let shift = "Morning";
```

```
switch (shift) {
```

```
    case "Morning":
```

```
        console.log("Shift: 06:00 - 14:00");
```

```
        break;
```

```
    case "Evening":
```

```
        console.log("Shift: 14:00 - 22:00");
```

```
        break;
```

```
    default:
```

```
        console.log("General Shift");
```

```
}
```

6. Loops

Loops are used to run the same code repeatedly as long as a condition is met.

* **for loop**: Used when you know how many times the code should run.

```
`for (let i = 0; i < 5; i++) { console.log(i); }`
```

* **while loop**: Used when you want the code to run as long as a condition is true.

```
`while (powerLevel < 100) { powerLevel++; }`
```

7. Functions

Functions are reusable blocks of code designed to perform a particular task.

```
``javascript
```

```
function calculatePowerOutput(voltage, current) {
```

```
    return voltage * current;
```

```
}
```

```
let result = calculatePowerOutput(220, 10); // Returns 2200
```

8. Arrays

Arrays are used to store multiple values in a single variable.let departments = ["HRDC", "Engineering", "Operations"];

```
console.log(departments[0]); // Output: HRDC
```

```
departments.push("Quality Control"); // Adds a new element
```

9. Events

Events are "things" that happen to HTML elements, which JavaScript can react to.

Examples include a button click, a mouse hover, or a page load.

```
<button  
onclick="showAlert()">Submit Data</button>
```

```
<script>
```

```
function showAlert() {
```

```
    alert("Data submitted successfully!");
```

```
}
```

```
</script>
```

10. DOM Manipulation Basics

The Document Object Model (DOM) is a programming interface for web documents. JavaScript can change the HTML and CSS of a page using the DOM.

- **document.getElementById(id):** Finds an element by its ID.
- **innerHTML:** Changes the content of an element.
- **style.property:** Changes the CSS of an element.

Example:

```
let element = document.getElementById("statusHeader");
```

```
element.innerHTML = "System Online";
```

```
element.style.color = "green";
```

Congratulations! You have completed Module 3.